AAYUSH KUMAR

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Education

Vellore Institute of Technology — **Vellore, Tamil Nadu**
Bachelor of Technology (B.Tech) in Computer Science and Engineering — *Jun 2024 – May 2028 (Expected)*

Technical Skills

Languages: Python, Go, JavaScript, SQL, C++
Frameworks & Libraries: FastAPI, Django, Flask, React.js, Next.js, Express.js, Node.js, LangGraph
Tools & Technologies: Docker, Kubernetes, Linux, PostgreSQL, MongoDB, Redis, RabbitMQ, S3, OCI, ffmpeg, Nginx, Git

Experience

Loopedin — **Dec 2025 – Present**
Software Engineer Intern — *Remote*

- Built and scaled AI-agent connectors using LangGraph, enabling structured multi-step workflows such as summarization, task orchestration, and automated document generation.
- Implemented concurrent execution of agent workflows and connector pipelines, reducing end-to-end AI response time by **25–35%**.
- Designed robust integration layers with optimized state management and execution flow, improving reliability and supporting high-throughput task processing.

Yuusoft — **Jul 2025 – Aug 2025**
Software Engineer Intern — *Remote, Singapore*

- Engineered an AI-assisted automation tool using Discord + Claude to modify code, generate patches, and push Git commits via natural language, significantly improving developer workflow efficiency.
- Built core backend and platform features for Yuusoft's visual novel creator tool, including scene-management logic, input-processing pipelines, and structured data flows.
- Improved platform performance and responsiveness by optimizing state management and render paths, reducing UI latency and unnecessary re-renders by 30% and improving overall UX reliability.

Projects

Rendition - Distributed Video Transcoding System | *Python, FastAPI, RabbitMQ, S3, ffmpeg, Next.js* | GitHub

- Built a distributed video transcoding platform that streams large browser uploads directly to S3-compatible storage via presigned multipart uploads and generates adaptive HLS renditions using horizontally scalable `ffmpeg` workers.
- Engineered a fault-tolerant job orchestration system with PostgreSQL, RabbitMQ, and the Outbox pattern, implementing worker leasing, heartbeats, stale-job recovery, retry backoff, and dead-letter handling to prevent lost or duplicate work under failures.
- Implemented source-aware encoding, master-playlist generation, and API-mediated playback delivery, alongside a Next.js dashboard providing upload progress tracking, job control, and HLS video playback.

Crate - Daemonless Container Runtime | *Go, Linux Namespaces, OCI Images* | GitHub

- Built a daemonless container runtime in Go supporting both rootless and rootful containers, with runtime state persisted directly on disk instead of a long-lived background daemon.
- Implemented container isolation using Linux namespaces (PID, UTS, Mount, User, Network), `pivot_root`-based filesystem setup, rootless networking via `pasta`, and PTY-backed interactive process execution.
- Developed OCI/Docker image support including registry pulls, manifest resolution, layer extraction, local blob/config storage, detached container lifecycle management (`start`, `stop`, `logs`, `ps`), and reference-based blob pruning.

Kargo - Open-Source PaaS on Kubernetes | *Express.js, Next.js, MongoDB, Docker, Kubernetes* | GitHub

- Developed core backend services for an open-source PaaS platform that automatically deploys user-provided Docker images onto Kubernetes, building secure APIs for authentication, deployment orchestration, and lifecycle management.
- Engineered automatic Kubernetes manifest generation (Deployments, Services, Ingress, volumes, secrets), eliminating manual cluster interaction and cutting deployment setup time by 60%.
- Improved deployment stability, scalability, and observability by adding container-status monitoring, error-resilient rollout logic, and health-check mechanisms to ensure a reliable and production-grade application.

Achievements & Extracurricular

- **Google Summer of Code 2025 @AboutCode:** Contributed to the open-source ecosystem by building a high-performance file explorer for ScanCode.io, improving navigation across large license, copyright, and vulnerability scan results.
- **Patent Published - Encrypted Traffic Intrusion Detection (2026):** Co-invented a multi-modal AI cybersecurity system for detecting threats in encrypted network traffic using metadata analysis and anomaly detection models.
- **Volunteer - Gravitas Website Team:** Helped build the backend for VIT's annual technical fest platform, serving **65k+ unique users** and handling **1M+ page visits**.
- **Winner - Cicada 3301 (CTF Competition):** Secured **1st place** in a university-wide cybersecurity challenge.
- **Third Place - Coderush 2.0 (Competitive Coding):** Achieved **3rd place** in a university-wide coding competition.